

# Remove Targets

from Array



target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: 0

*Check if* **current element** **!= target**

target = 6

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| <b>2</b> | <b>2</b> | <b>5</b> | <b>6</b> | <b>8</b> |
|----------|----------|----------|----------|----------|

Next element index: **1**

*Increment next element index*

target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: **1**

*Check if* **current element** **!= target**

target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: **2**

*Increment next element index*

**target = 6**

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| <b>2</b> | <b>2</b> | <b>5</b> | <b>6</b> | <b>8</b> |
|----------|----------|----------|----------|----------|

Next element index: **2**

*Check if* **current element != target**

target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: 3

*Increment next element index*

target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: 3

*Check if* **current element**  $\neq$  target



target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: 3

*Check if* **current element**  $\neq$  target

target = 6

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 2 | 5 | 6 | 8 |
|---|---|---|---|---|

Next element index: 4

*Increment next element index*

Next element index: **4**

Time complexity  **$O(N)$**

Space complexity  **$O(1)$**